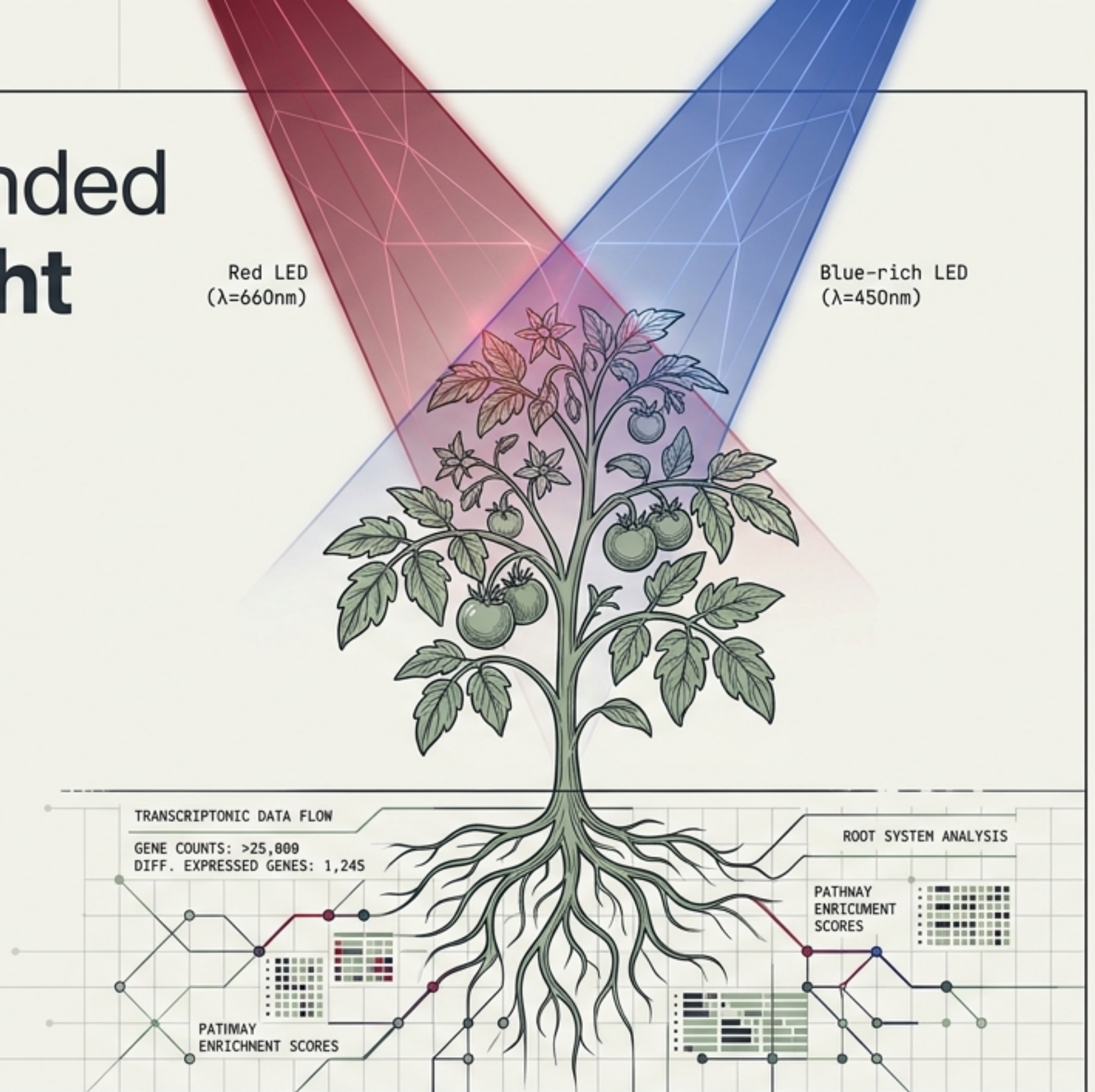


Decoding the unintended consequences of **light design** in space agriculture.

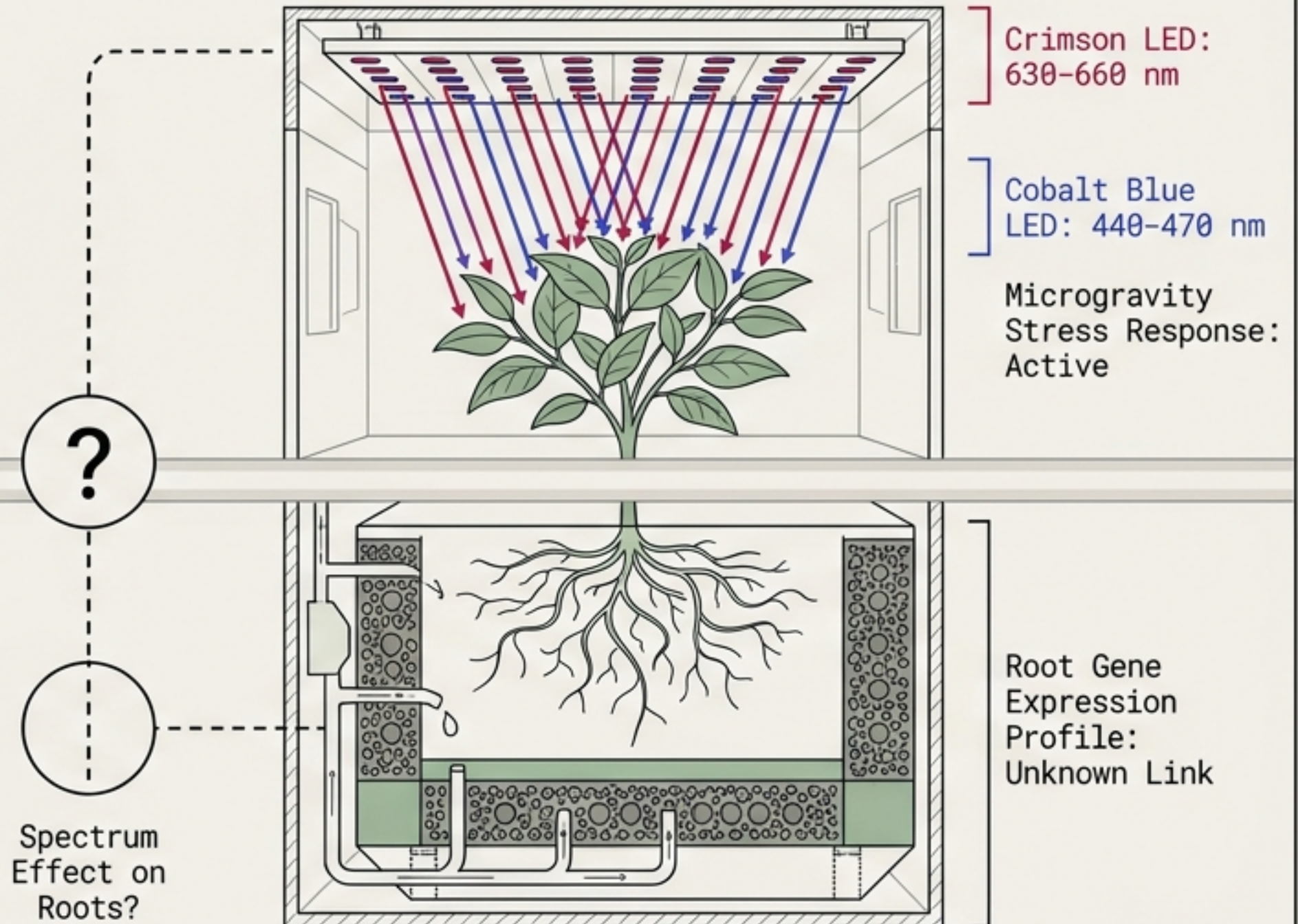
A PhysioSpace analysis of the NASA VEG-05 / OSD-767 Tomato Experiment.



Solving a problem at the top of the plant creates invisible stress at the bottom.

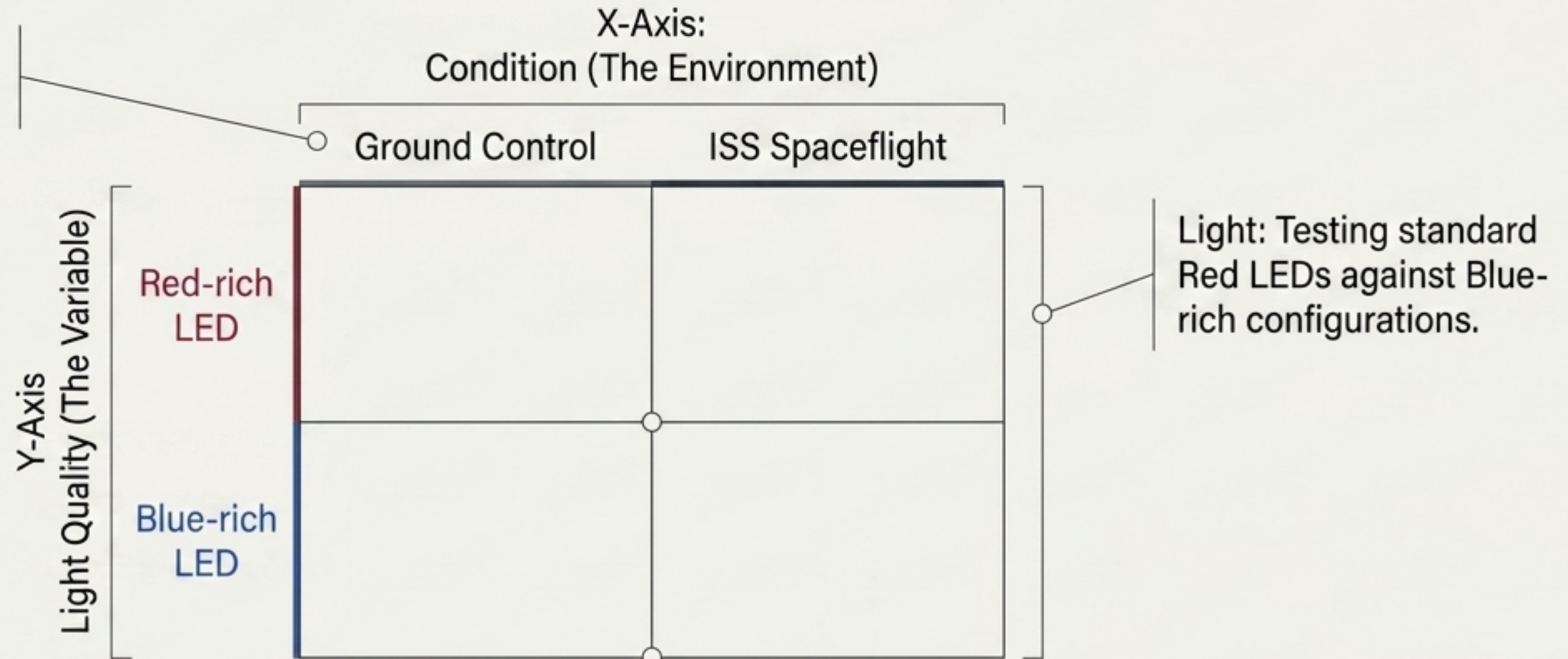
To grow food in space, engineers use specialized red- and blue-rich LEDs to drive photosynthesis.

But a plant is an interconnected system. The VEG-05 experiment reveals that the light spectrum chosen for the leaves dramatically alters how the roots experience the stress of microgravity.



The VEG-05 experiment tests microgravity against two distinct light spectra.

Condition: Isolating the core microgravity footprint.



Interaction: Analyzing DESeq2 models (~ light + condition + light:condition) to find where light design modulates spaceflight stress.



Diagnostic Legend

Leaves respond primarily to light, while roots respond primarily to spaceflight.



Organ: Canopy Leaves

Organ: Adventitious Roots

Dominant Driver: Light Spectrum

Dominant Driver: Microgravity (Condition)

DEG Count: 2,314 Light vs. 527 Condition

DEG Count: 704 Condition vs. 256 Light

Primary State: Antenna proteins and photosynthesis maintained.

Primary State: Severe oxidative stress and phenylpropanoid metabolism activation.

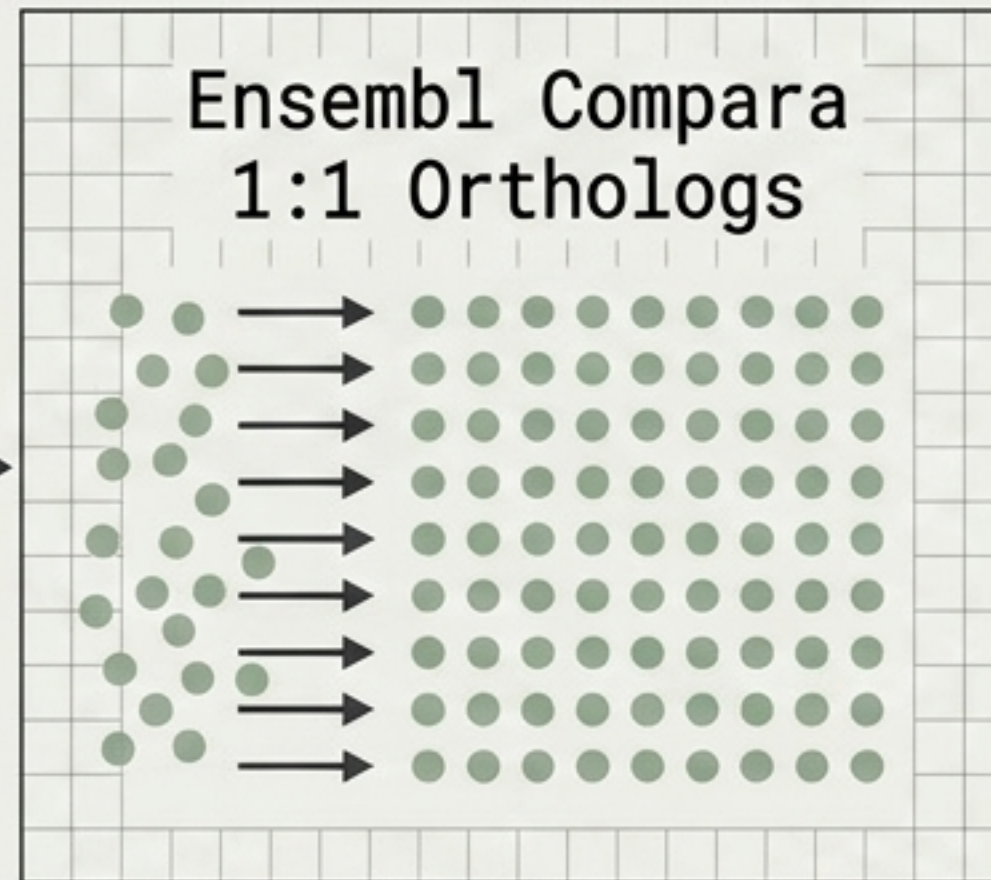
PhysioSpace translates hidden tomato transcriptomes into known stress patterns.

1. Raw Data



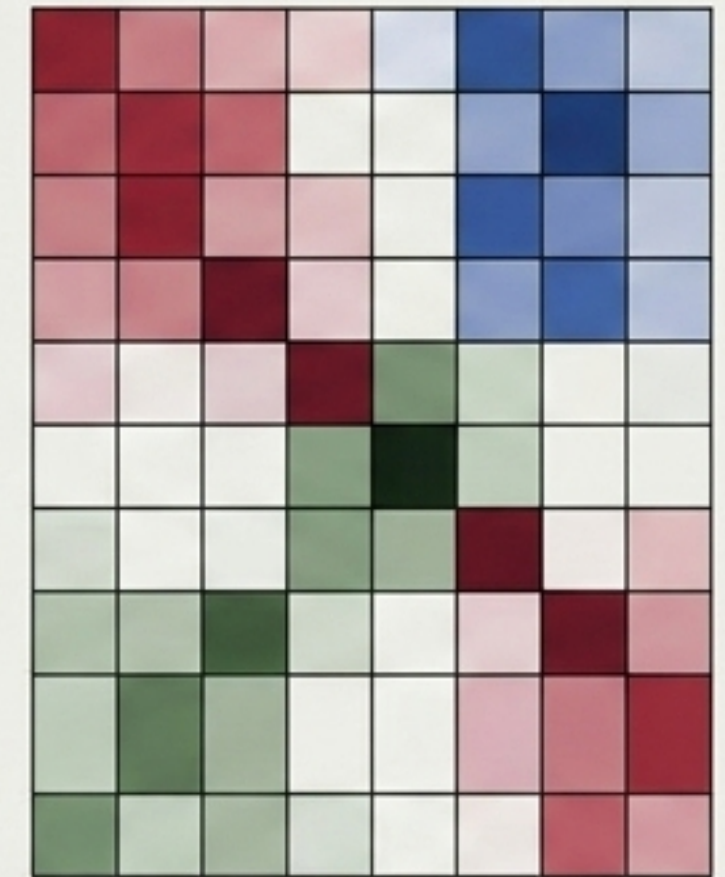
Extract complex spaceflight gene expression from the dwarf tomato.

2. Translation Engine



Map the data to *Arabidopsis thaliana* orthologs to use as a universal biological translator.

3. Pattern Recognition



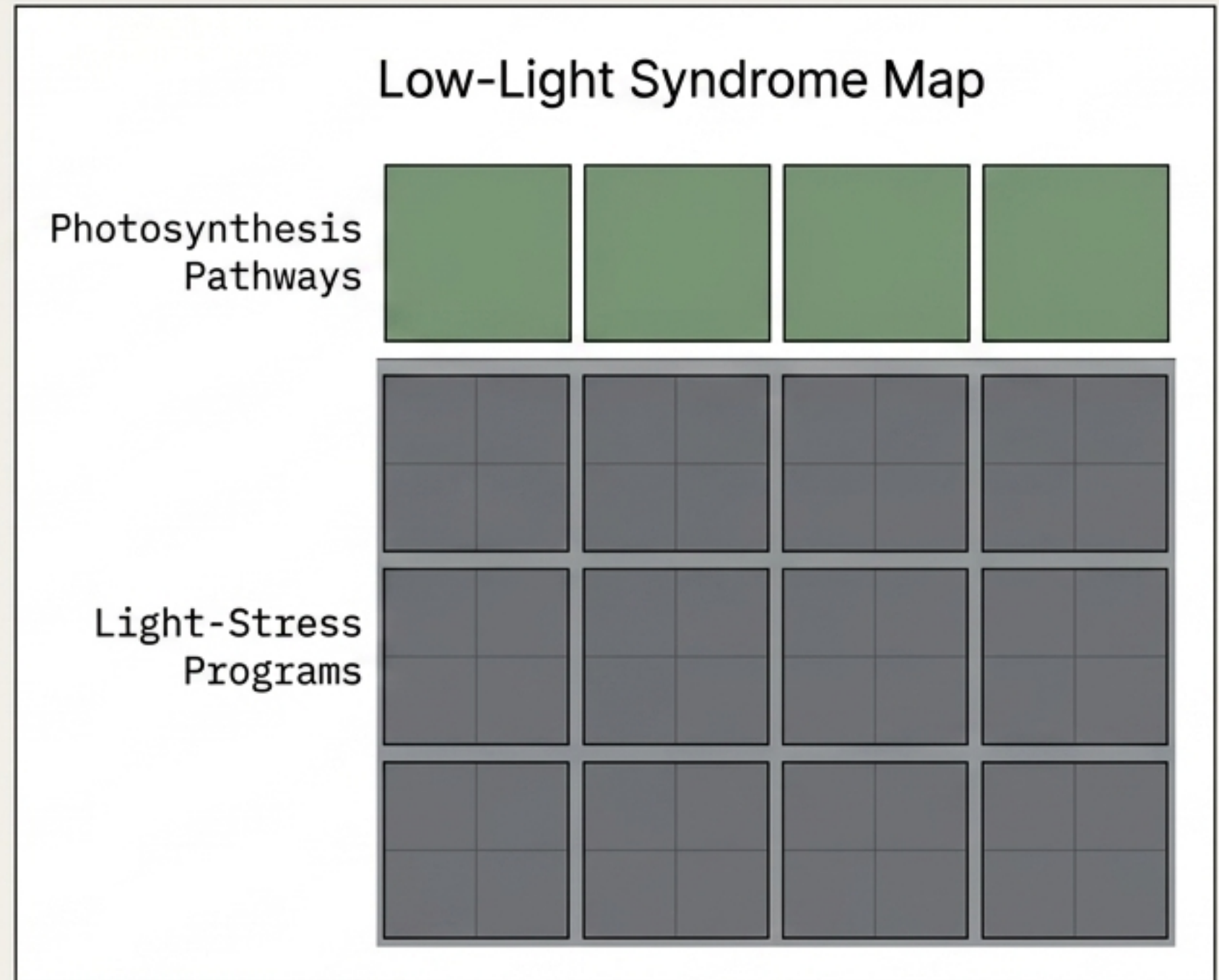
Project data onto pre-built reference compendia to identify which established stress programs (e.g., hormone, biotic, oxidative) the plant is activating.

Leaves in space exhibit a suppressed low-light syndrome signature.

	Environments	
	Ground Control	Space-flight
Conditions	Low Light	Bright Light

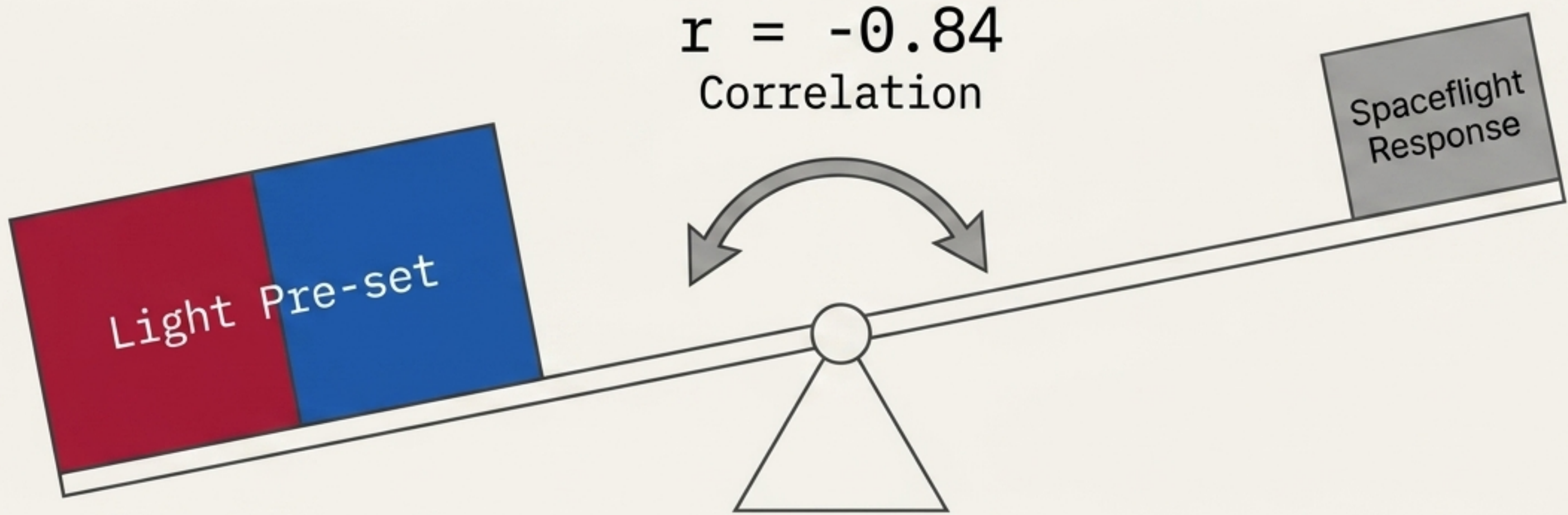
Key Finding: Despite constant LED illumination, PhysioSpace decoding reveals that spaceflight leaves suppress their typical light-stress programs.

Insight: The microgravity environment fundamentally alters how the canopy processes photons, creating a molecular footprint that mimics low-light conditions even under bright LEDs.



Pathways maxed out by light have no capacity left to respond to spaceflight.

		Environments	
		Ground Control	Spaceflight
Conditions	Low Light		
	Bright Light		

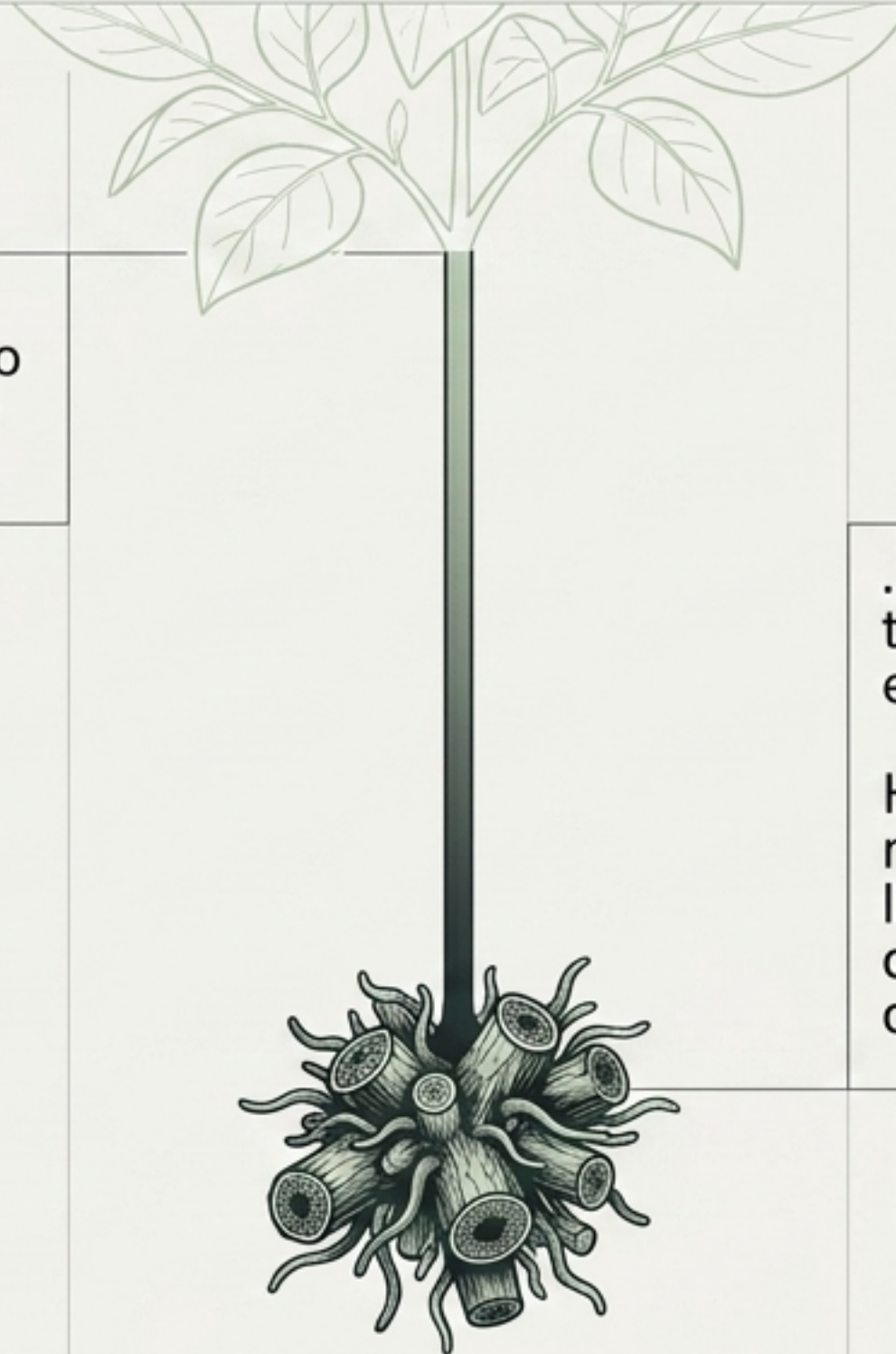


Mechanism (Compensatory Rescue): If a specific light quality (Red or Blue) strongly downregulates or upregulates a pathway on the ground, that pathway physically cannot shift much further when subjected to spaceflight. The light acts as a biological ceiling.

Beneath the surface, microgravity triggers a severe, localized stress response.

Environments	Conditions
Ground Control	Spaceflight
Low Light	Bright Light

While the leaves utilize compensatory dynamics to maintain antenna proteins and photosynthesis...

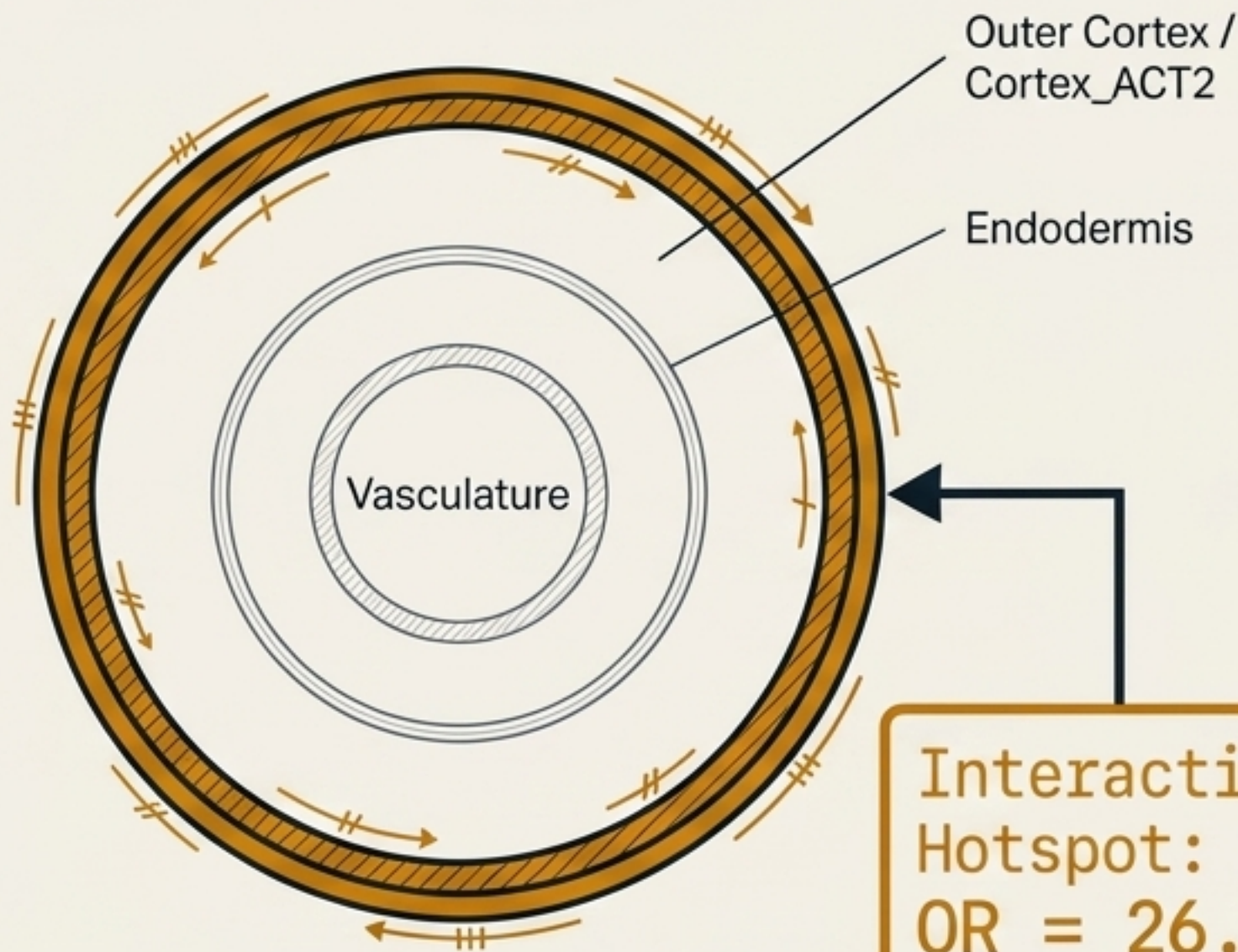


...the adventitious roots bear the brunt of the spaceflight environment.

Here, the condition of microgravity overrides the light signals, acting as the driver of genetic disruption.

Spaceflight stress is radially asymmetric, concentrating in the outer root cortex.

Asymmetry Target



Interaction
Hotspot:
OR = 26.9

Data Point: Cell-type deconvolution reveals the stress is not uniform.

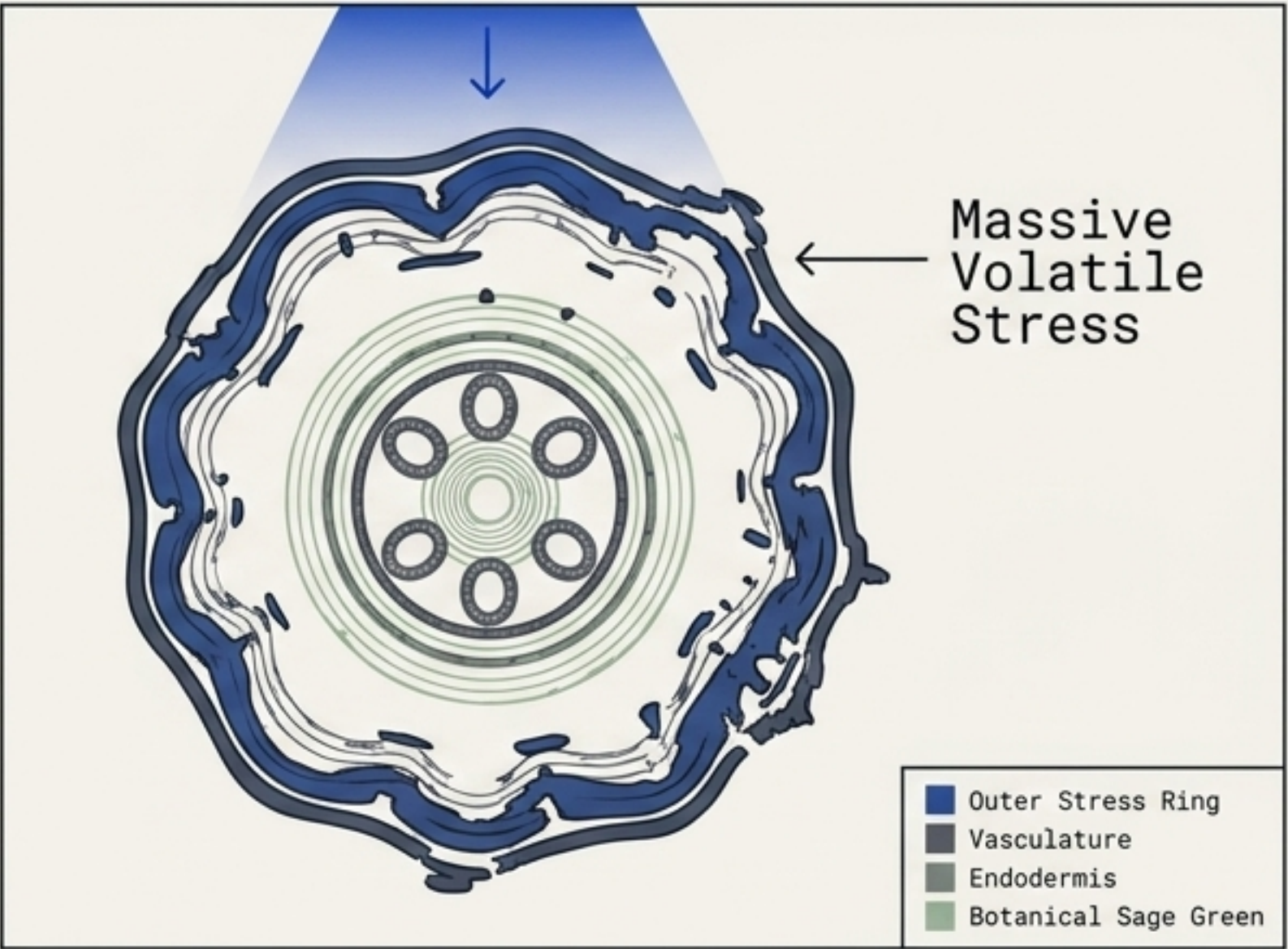
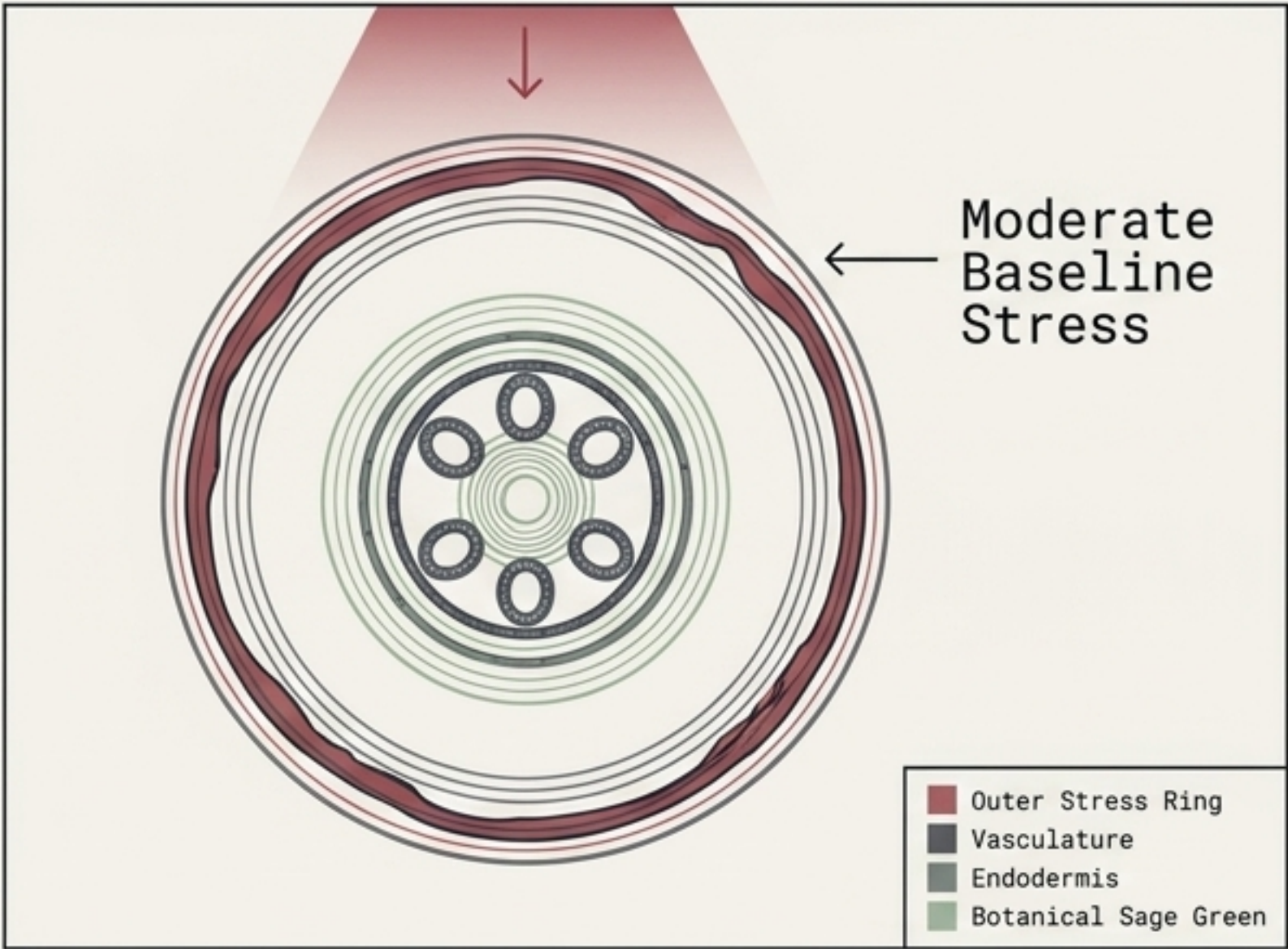
- The outer cortex experiences massive transcriptomic disruption, identifying it as the primary interaction hotspot (Odds Ratio of 26.9).
- The inner layers remain largely insulated and balanced.

					
 Outer Cortex	0.3	28.0	4.8	8.0	10.0
 Vasculature	0.0	1.0	3.3	10.0	4.9
 Endodermis	0.0	0.0	0.1	1.0	3.1

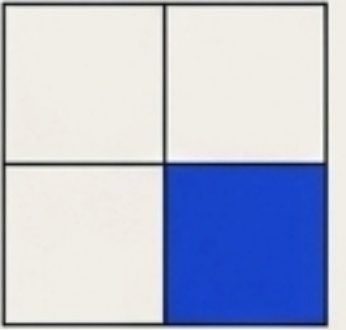
The chosen canopy light spectrum dictates the severity of the root's spaceflight stress.

Environments	Conditions	Conditions
Ground Control		
Spaceflight		
Low Light		
Bright Light		

The roots are in total darkness. Yet, systemic signals sent from the leaves mean the roots experience microgravity entirely differently depending on the overhead LEDs.



Blue light hyperactivates biotic and hormone defense signaling in the roots.



Blue Light /
Spaceflight

Spaceflight Control Baseline (+58.4)

The Data:

The Flight × Light interaction creates a massive modulatory signal under Blue LEDs. This -44.2 shift is the single largest interaction effect at any scale in the study.

Result: Blue light essentially throws the root's defense mechanisms into overdrive.

Shift: -44.2 (Biotic.Hormone)

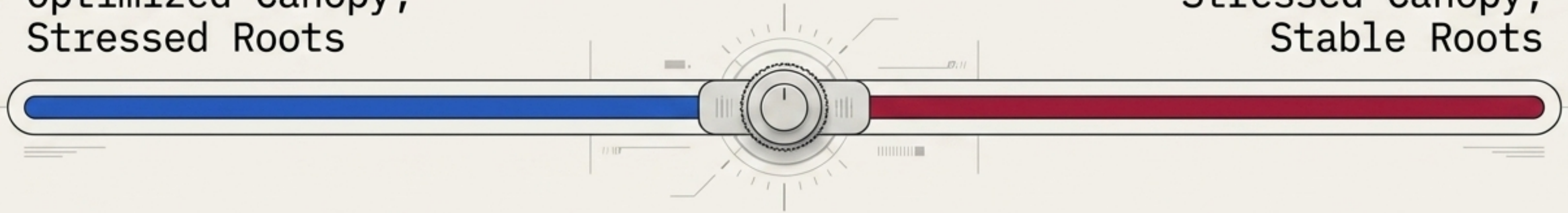
Every lighting choice forces a biological compromise.

	Red LED Impact	Blue-rich LED Impact
Leaf Ecosystem	Amplifies hormone stress and MAPK signaling.	Enhances induction of phenylpropanoid biosynthesis (photosynthesis maintained)
Root Ecosystem	Maintains a moderate, baseline spaceflight stress response.	Hyperactivates biotic/hormone defense with extreme cortex asymmetry.

The Light-Design Trade-off: You cannot optimize the plant in isolation.

Optimized Canopy,
Stressed Roots

Stressed Canopy,
Stable Roots



The ISS lighting environment acts as a master biological switch. Blue-rich light successfully solves the engineering problem of maintaining leaf photosynthesis in space, but it inadvertently acts as a systemic trigger, causing massive, asymmetric defense cascades in the root zone.

Optimizing space agriculture requires designing for the whole ecosystem.



The VEG-05 PhysioSpace decoding proves that in the extreme environment of microgravity, localized interventions have systemic consequences.

Future spaceflight agriculture must move beyond maximizing canopy yield, developing lighting formulas that balance photosynthesis with root-zone stability to ensure true multi-generational crop viability.